Week 2, Lecture 3
Photodetection

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See Chapter 7.2 of Bachor, Hans-A. / Ralph, Timothy C. A guide to experiments in quantum optics (2019)

See Chapter 5.2 of Leonhardt, U. Essential Quantum Optics From Quantum Measurements to Black Holes (2010)

Silberhorn, C. (2007). Detecting quantum light. Contemporary Physics

Review

What you learned last lecture…

Density operators

To describe mixed ensembles of quantum states, we use density operators:

$$\hat{\rho} = \sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|$$

p_i: probability of being in state $|\psi_i\rangle$

Thermal states of light

$$\hat{\rho} = \frac{1}{\bar{n}+1} \sum_n \left(\frac{\bar{n}}{\bar{n}+1}\right)^n \left|n\right\rangle \left\langle n\right|$$

Photon statistics

- **Super-poissonian**

$$\mathrm{var}\,(n) > \bar{n}$$

  - e.g. thermal state $\hat{\rho}_{\mathrm{th}}$

- **Poissonian**

$$\mathrm{var}\,(n) = \bar{n}$$

  - e.g. coherent state $\left|\alpha\right\rangle$

- **Sub-poissonian**

$$\mathrm{var}\,(n) < \bar{n}$$

  - e.g. photon number state $\left|n\right\rangle$

Super-poissonian statistics: can be explained by a semi-classical model (quantized atom and classical light wave)

Sub-poissonian statistics: no semiclassical explanation, provides a clear demonstration that it is light itself that is quantized

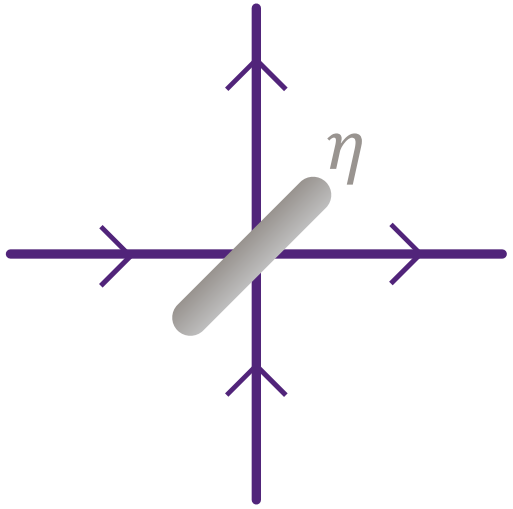
Review

What you learned last lecture…

Beam splitters transform the mode operators as:

$$\hat{a}_1' = \sqrt{\eta}\,\hat{a}_1 - \sqrt{1-\eta}\,\hat{a}_2$$

$$\hat{a}_2' = \sqrt{1-\eta}\,\hat{a}_1 + \sqrt{\eta}\,\hat{a}_2$$



Splitting n photons:

$$|n\rangle_1 |0\rangle_2 \rightarrow \sum_{k=0}^{n} \sqrt{\binom{n}{k}} t^k r^{n-k} |k\rangle_1 |n-k\rangle_2$$

Beam splitters can be used to model optical loss

Review – Measurements in quantum mechanics

- Measurements are represented by Hermitian operators
- These operators are called *observables*
- Hermitian $\Longrightarrow$ Eigenvalues are real
- Measurement outcome: eigenvalues of the observable
- When an observable is measured, the state of the system is projected onto the eigenstate corresponding to the measured eigenvalue
- For the observable $\hat{O}$, we have the eigenvalue equation:

$$\hat{O}\,|\psi_\lambda\rangle = \lambda\,|\psi_\lambda\rangle$$

$|\psi_\lambda\rangle$:state that gives outcome λ with certainty

Measurement in quantum optics

- For a single simple harmonic oscillator, the operator $\hat{n}$ is an observable

$$\hat{n} = \hat{a}^\dagger \hat{a}$$

- Eigenvalues are non-negative integers n
- Eigenvectors are Fock states $|n\rangle$
- Eigenvalue equation: $\hat{n}\,|n\rangle = n\,|n\rangle$

How do we measure $\hat{n}$?

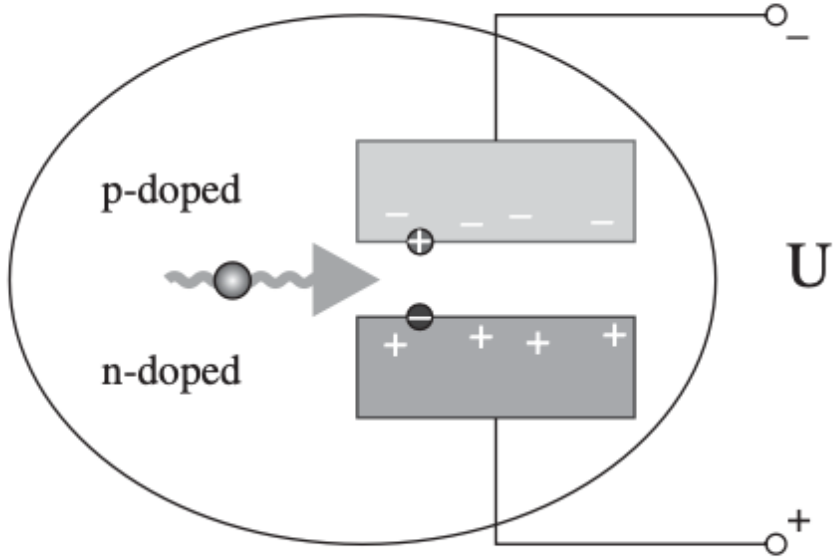
- Techniques are different for bright light eg. $|\alpha\rangle$ vs. low intensity eg. single photons $|1\rangle$
- Through measuring $\hat{n}$, we can also make measurements on the quadrature operators $\hat{x}$ and $\hat{p}$

Useful resources

- See Chapter 7.2 of Bachor, Hans-A. / Ralph, Timothy C. A guide to experiments in quantum optics (2019)
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- Silberhorn, C. (2007). Detecting quantum light. Contemporary Physics

Photodetection

- **Most photodetectors apply a version of the photoelectric effect to operate**
- Radiation ionizes a piece of photosensitive material and produces freely moving electrons
- This produces an electric current (photocurrent) that can be amplified and handled by electronic means

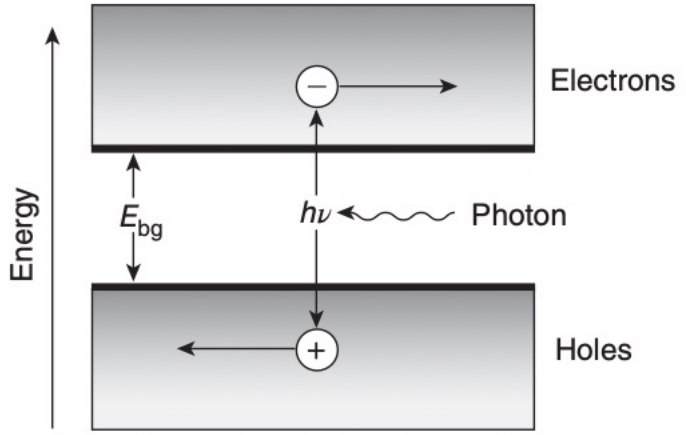


Silberhorn, C. (2007). Detecting quantum light. Contemporary Physics, 48(3), 143–156

- Conventional photodiodes: A photon creates an electron-hole pair in the region between the p-doped and n-doped layers such that a current can flow. This photocurrent is proportional to the photon flux.
- **Because of noise from thermal excitations, conventional photodiodes cannot be used for the detection of single photons**

Avalanche photodiodes

- **Single photons can be detected through avalanche photodiodes (APD)**
- A single electron is typically liberated by a single photon
- High voltage is applied across a so-called ‘multiplication region’ of the diode
- Charge carriers are amplified to macroscopic current pulses.
- **A single photon triggers an avalanche of charge carriers**

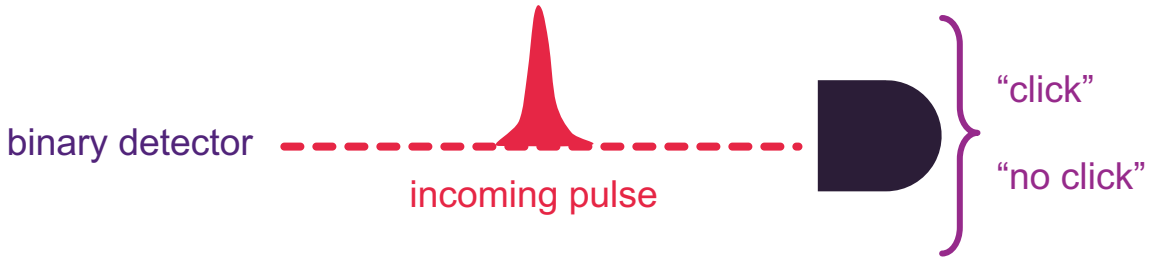


p. 242 Bachor, Hans-A. / Ralph, Timothy C.
A guide to experiments in quantum optics (2019)

Avalanche photodiodes

- Avalanche photodiodes are sensitive enough to detect single photons
- But they are oversensitive for more than one photon – *binary detector, on-off detector*

$$\Pi_0 = |0\rangle\langle 0| \qquad \Pi_1 = \mathbb{1} - |0\rangle\langle 0|$$

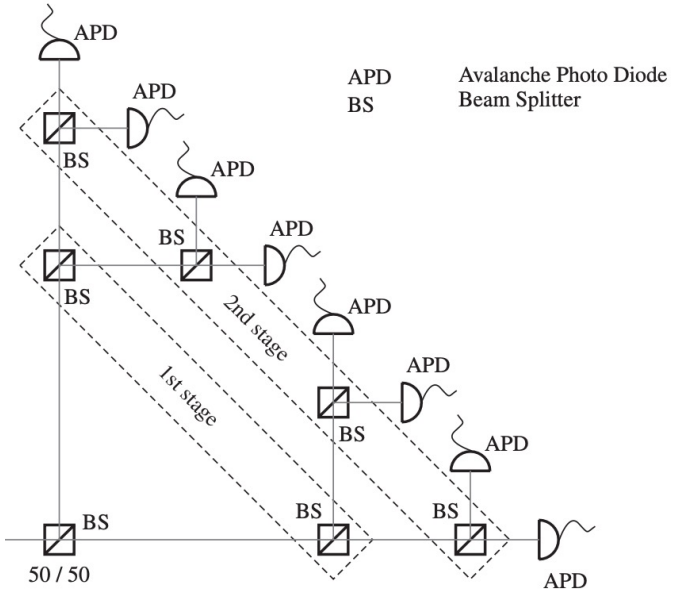


- APDs are not *photon number resolving* detectors (PNR)

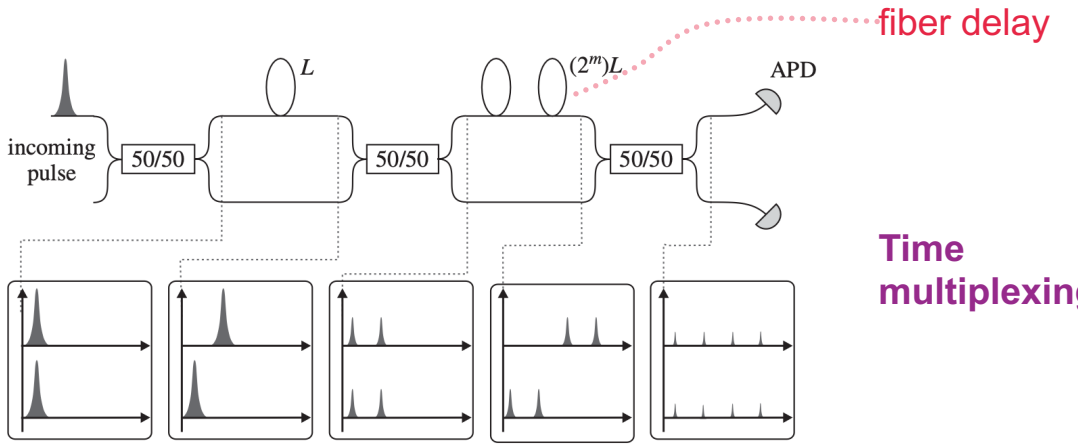
Avalanche photodiodes

How do we measure $\hat{n}$ using APDs?

- Multiplexing with Avalanche photo diodes (APD): beam splitters distribute the light on many APDs
- If the light is sufficiently diluted such that each photodiode will hardly encounter more than one photon, the multiplexed detector measures the accurate photon number
- **Time multiplexing:** light is split and delayed through fibre loops
- Only need two APDs to resolve photon number



Spatial multiplexing



Time multiplexing

p. 108 Leonhardt, U. Essential Quantum Optics From Quantum Measurements to Black Holes (2010)

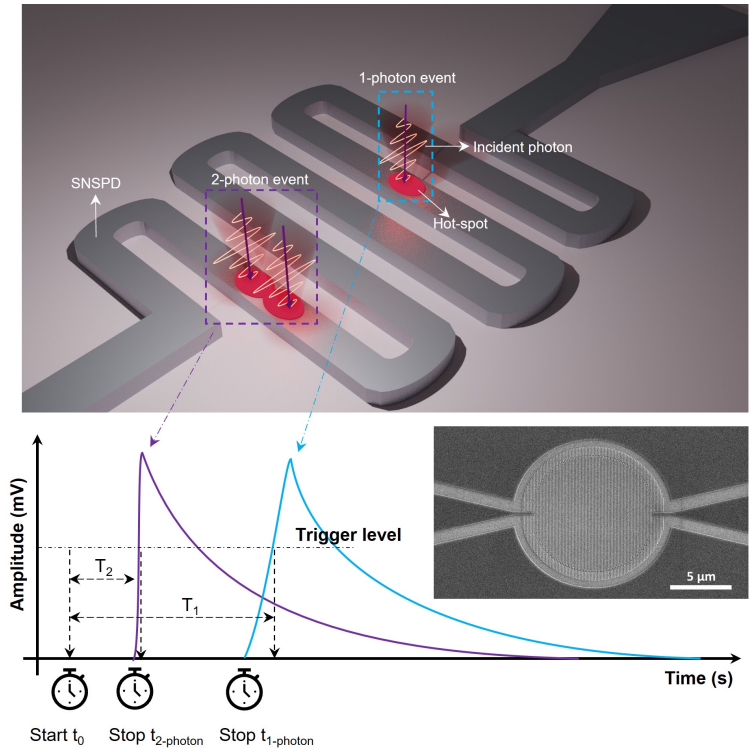
Photon number resolving detectors

- **Ideal photon number resolving detectors implement projection on individual Fock states**

$$\Pi_n = |n\rangle\langle n|$$

- **Need physical processes that have a measurable response to extremely low changes in intensity of incident light**

- Superconducting-nanowire single-photon detectors (SNSPD) are used as photon number resolving detectors
- A photon is absorbed in a superconducting nanowire, the photon energy is transferred to electrons and phonons
- This energy produces a small non-superconducting region known as a “hot spot” and the wire transitions to a resistive state
- This blocks the superconducting current resulting in a voltage spike, with appropriate signal processing: photon number resolution can be achieved.



J. W. Niels Los, et. al. High-performance photon number resolving detectors for 850–950 nm wavelength range. APL Photonics 1 June 2024; 9 (6): 066101.

Can resolve up to 7 photons

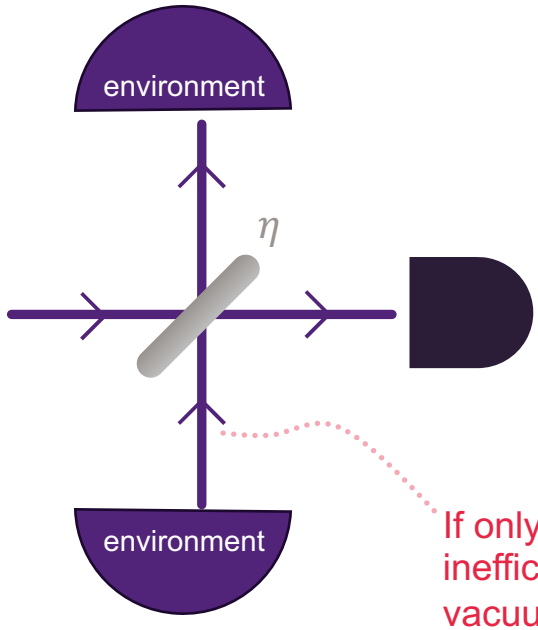
Realistic detectors

Detection inefficiency

- Not all incident photons create charge carriers
- Realistic photodiodes work with a detector efficiency η
- η is defined as the ratio between detected photons and incident photons
- Can be modelled by a beam splitter of transmissivity η immediately before the detector

Dark counts

- False detection events corresponding to the noise of the detector
- Dark count rate is defined by the number of detection events in the absence of an optical signal



If only modelling detector inefficiency:
vacuum $|0\rangle_E$

For dark counts:
thermal state $\hat{\rho}_{\mathrm{th}}$

Homodyne detection

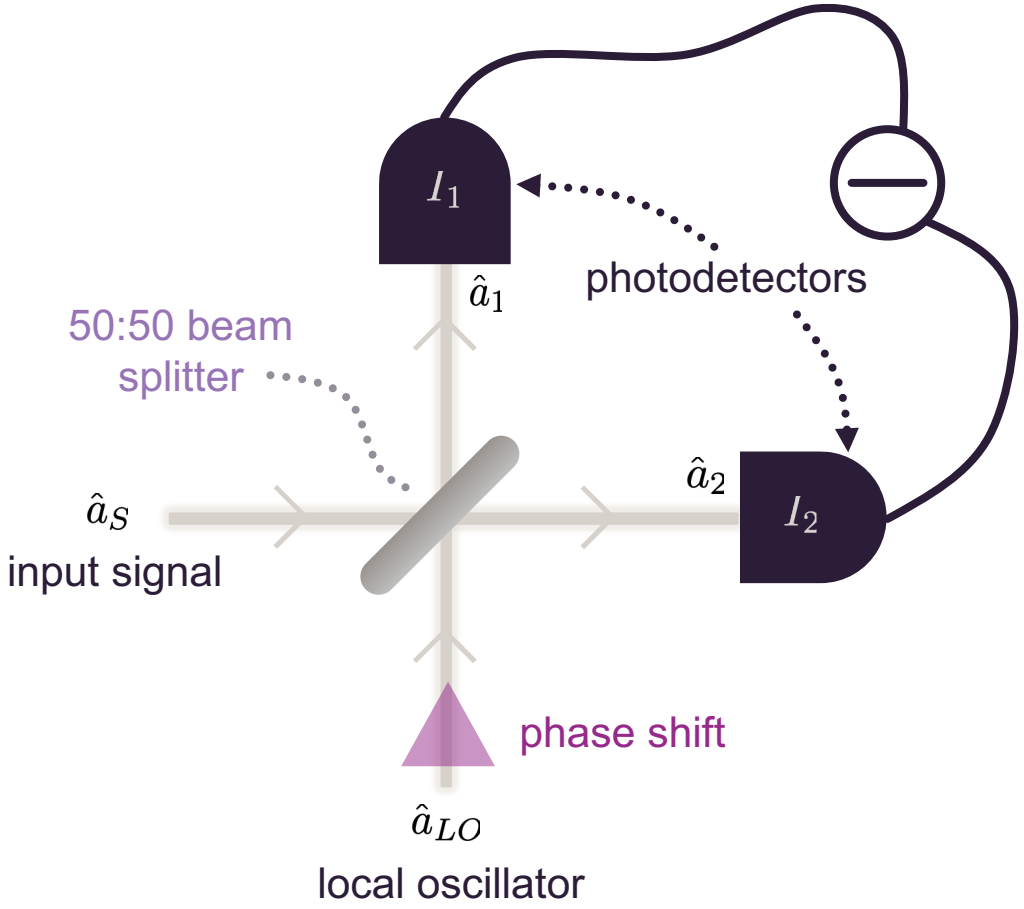
Homodyne detection is used to measure the quadrature variables $\hat{x}$ and $\hat{p}$

$$\hat{x} = \left(\hat{a}^\dagger + \hat{a}\right)$$
$$\hat{p} = i\left(\hat{a}^\dagger - \hat{a}\right)$$

Can be used on states with low photon number $|\alpha|^2$

Need to use the interference between input signal and an intense laser (local oscillator)

Local oscillator must be phase locked to the input



Phase shift

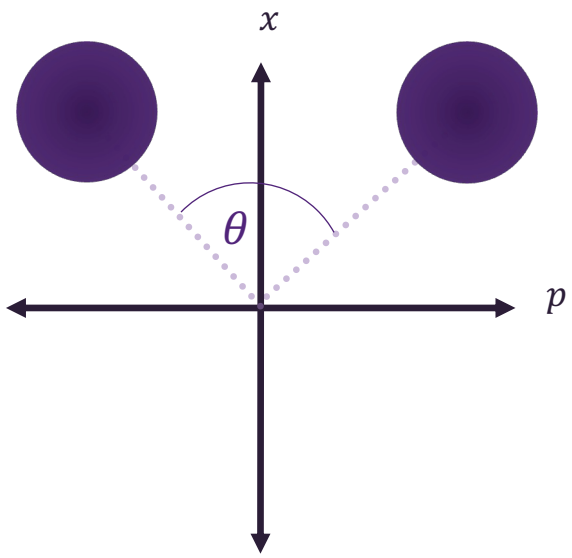
Unitary operator describing phase shift:

$$\hat{U}(\theta) = \exp(-i\theta\hat{n})$$

$$\hat{n} = \hat{a}^{\dagger}\hat{a}$$

The phase shift operator performs rotations in phase space

$$\hat{U}(\theta)|\alpha\rangle = |\alpha e^{-i\theta}\rangle$$



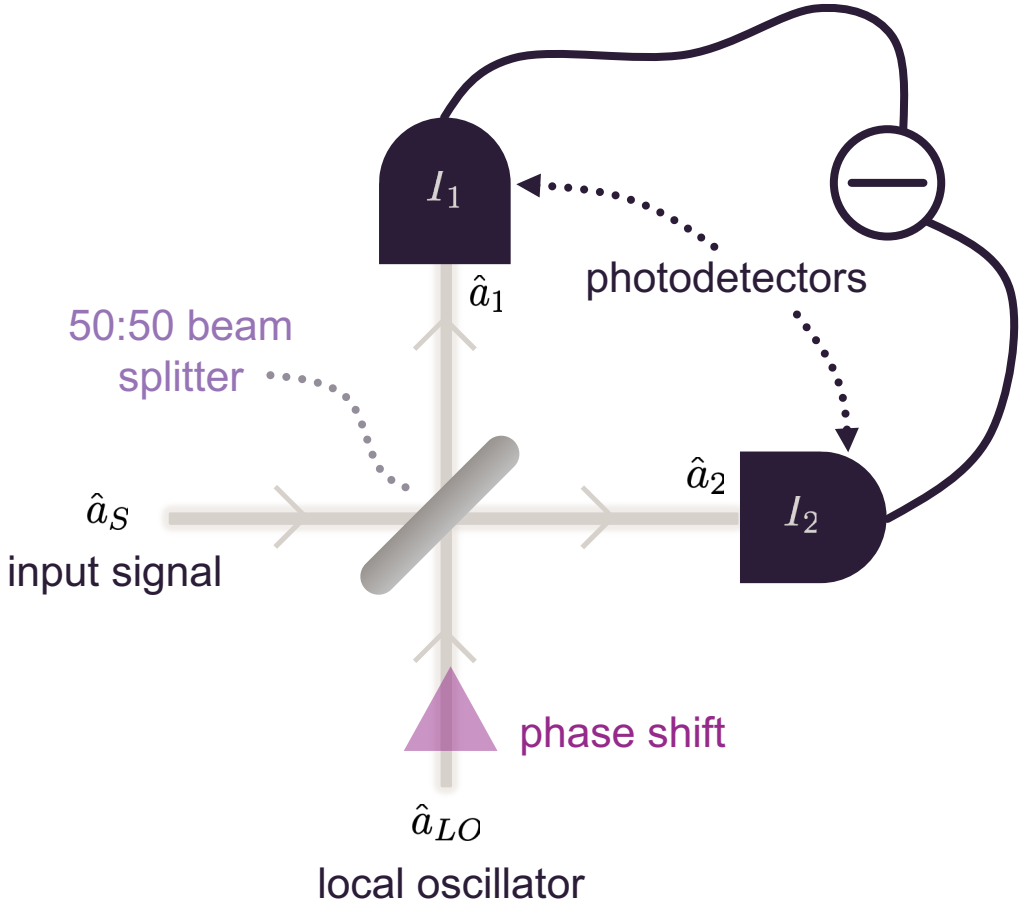
Homodyne detection

Combine input signal and local oscillator at 50:50 beam splitter

Beamsplitter transformation

$$\hat{a}_1 = \frac{\hat{a}_s + \hat{a}_{LO}}{\sqrt{2}}$$

$$\hat{a}_2 = \frac{\hat{a}_s - \hat{a}_{LO}}{\sqrt{2}}$$



Homodyne detection

Measure the photocurrents

$$I_1 \propto \hat{n}_1 = \hat{a}_1^\dagger \hat{a}_1$$
$$I_2 \propto \hat{n}_2 = \hat{a}_2^\dagger \hat{a}_2$$

Subtract the photocurrents

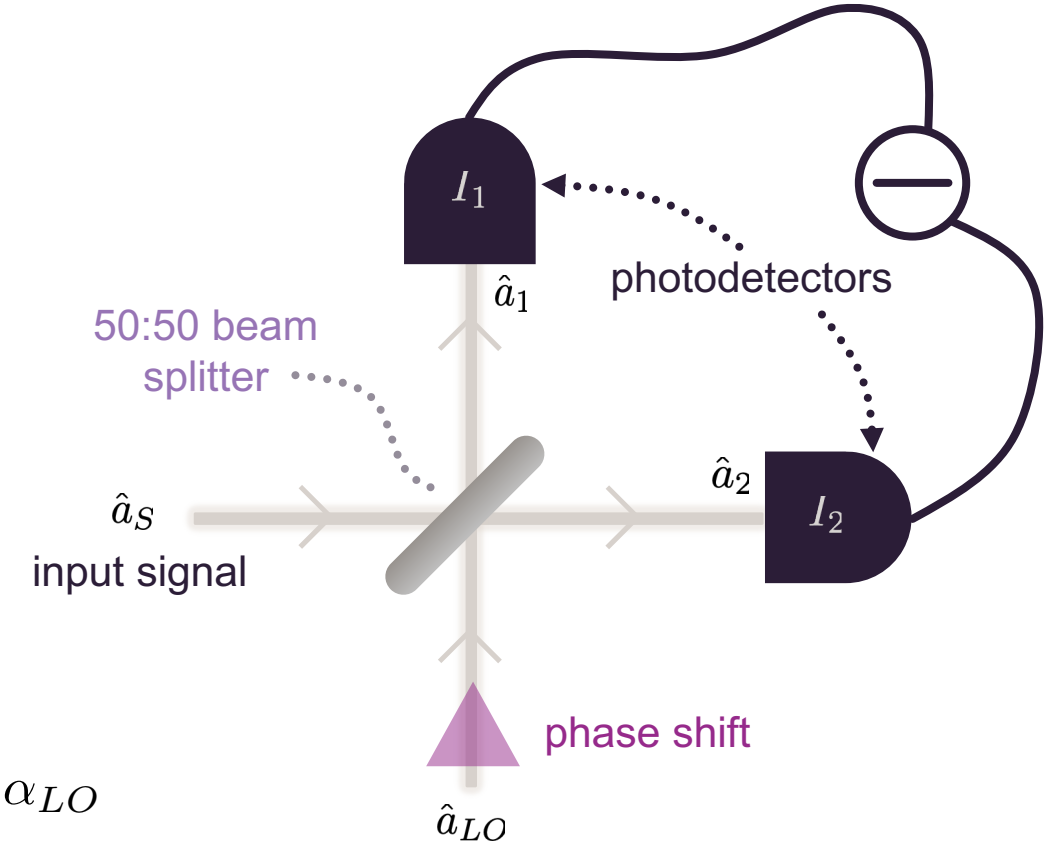
$$I_{12} = I_1 - I_2 \propto \hat{n}_1 - \hat{n}_2$$

$$\hat{n}_1 - \hat{n}_2 = \frac{1}{2}\left(\hat{a}_S^\dagger + \alpha_{LO}^*\right)\left(\hat{a}_S + \alpha_{LO}\right) - \frac{1}{2}\left(\hat{a}_S^\dagger - \alpha_{LO}^*\right)\left(\hat{a}_S - \alpha_{LO}\right)$$

local oscillator is a high intenstiy beam, we can treat it classically $\hat{a}_{LO} \to \alpha_{LO}$.

$$= \frac{1}{2}\left[\hat{a}_S^\dagger \hat{a}_S + |\alpha_{LO}|^2 + \hat{a}_S \alpha_{LO}^* + \hat{a}_S^\dagger \alpha_{LO} - \left(\hat{a}_S^\dagger \hat{a}_S + |\alpha_{LO}|^2 - \hat{a}_S \alpha_{LO}^* - \hat{a}_S^\dagger \alpha_{LO}\right)\right]$$

$$= \hat{a}_S \alpha_{LO}^* + \hat{a}_S^\dagger \alpha_{LO}$$

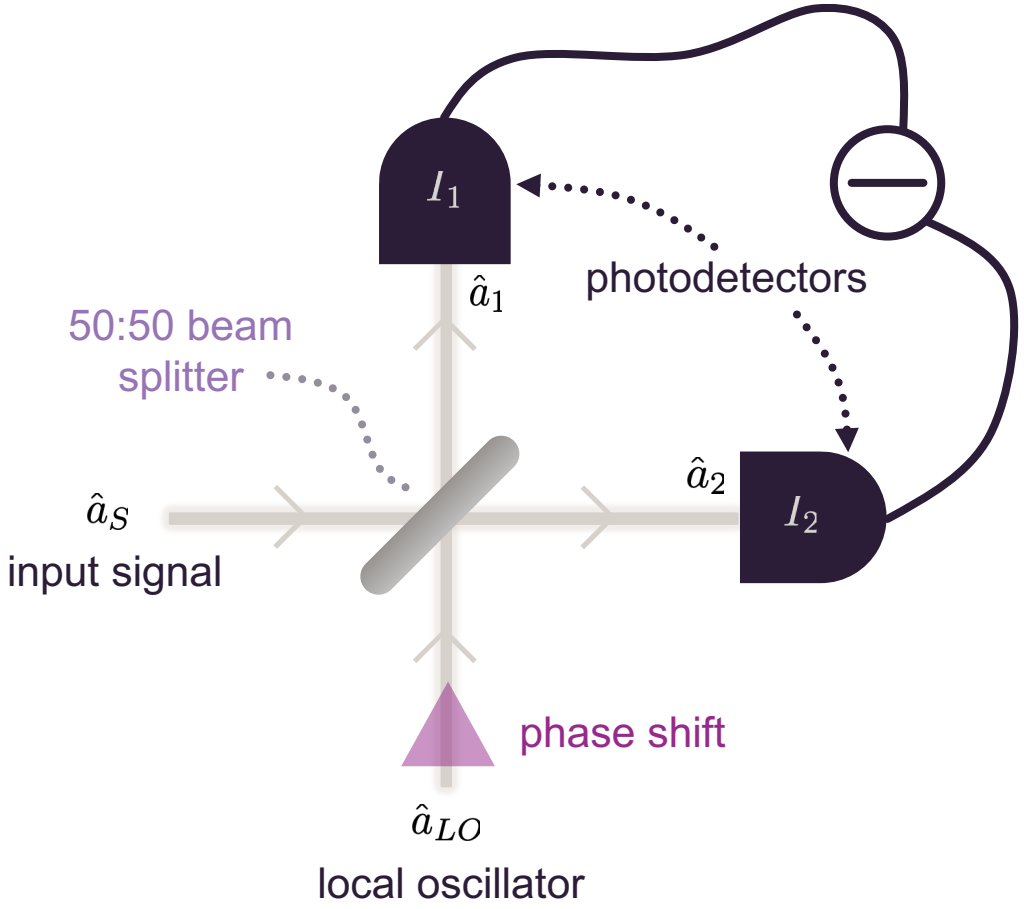


Homodyne detection

$$\hat{x} = \left(\hat{a}^\dagger + \hat{a}\right) \qquad \hat{a} = \frac{1}{2}\left(\hat{x} + i\hat{p}\right)$$
$$\hat{p} = i\left(\hat{a}^\dagger - \hat{a}\right) \qquad \hat{a}^\dagger = \frac{1}{2}\left(\hat{x} - i\hat{p}\right)$$

$$\hat{a}_S = \frac{1}{2}\left(\hat{x} + i\hat{p}\right) \qquad \alpha_{LO} = e^{i\phi}\left|\alpha_{LO}\right|$$

$$\begin{aligned} I_{21} \propto \hat{n}_2 - \hat{n}_1 &= \hat{a}_S\alpha_{LO}^* + \hat{a}_S^\dagger\alpha_{LO} \\ &= \frac{1}{2}\left(\hat{x} + i\hat{p}\right)\left|\alpha_{LO}\right| e^{-i\phi} + \frac{1}{2}\left(\hat{x} - i\hat{p}\right)\left|\alpha_{LO}\right| e^{i\phi} \\ &= \frac{1}{2}\left(\hat{x} + i\hat{p}\right)\left|\alpha_{LO}\right|\left(\cos\phi - i\sin\phi\right) \\ &\quad + \frac{1}{2}\left(\hat{x} - i\hat{p}\right)\left|\alpha_{LO}\right|\left(\cos\phi + i\sin\phi\right) \\ &= \left|\alpha_{LO}\right|\left(\cos\phi\,\hat{x} + \sin\phi\,\hat{p}\right) \end{aligned}$$



- Depending on our choice of the phase of the local oscillator ϕ, we can measure either $\hat{x}$ or $\hat{p}$.
- The phase of the local oscillator can be changed with a phase shift.

Summary

What we covered this week…

States

Fock photon number states

$$\{|n\rangle\} = \{|0\rangle, |1\rangle, |2\rangle, \cdots\}$$

Coherent states

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

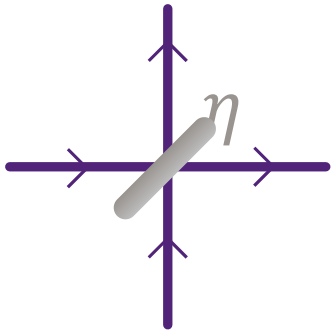
Thermal states

$$\hat{\rho} = \frac{1}{\bar{n}+1} \sum_n \left(\frac{\bar{n}}{\bar{n}+1}\right)^n |n\rangle\langle n|$$

Operations

Beam splitters

$$\hat{a}_1' = \sqrt{\eta}\,\hat{a}_1 - \sqrt{1-\eta}\,\hat{a}_2$$
$$\hat{a}_2' = \sqrt{1-\eta}\,\hat{a}_1 + \sqrt{\eta}\,\hat{a}_2$$



Phase shift

$$\hat{U}(\theta) = \exp(-i\theta\hat{n})$$

Measurement

- Avalanche photodiodes are sensitive enough to detect single photons
- Homodyne detection is used to measure quadratures $\hat{x}$ and $\hat{p}$

